

Robust and Efficient Prompt Rehearsal-Free Federated Continual Learning

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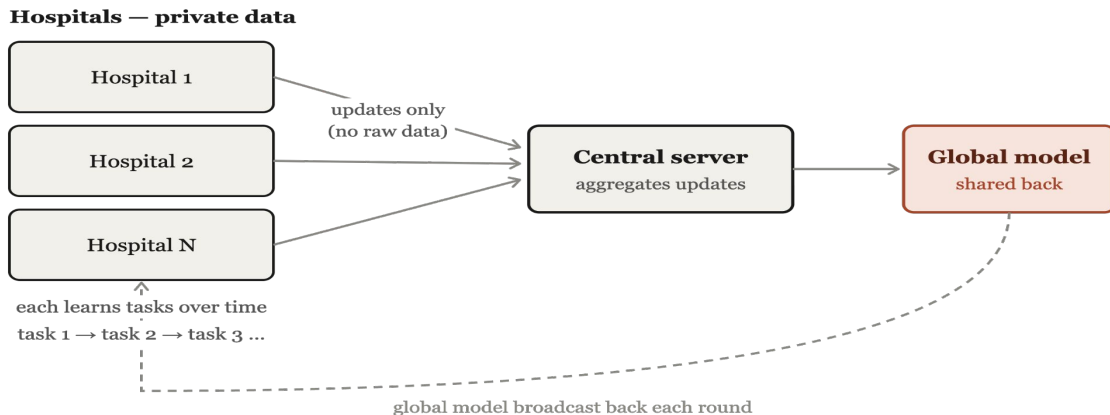
భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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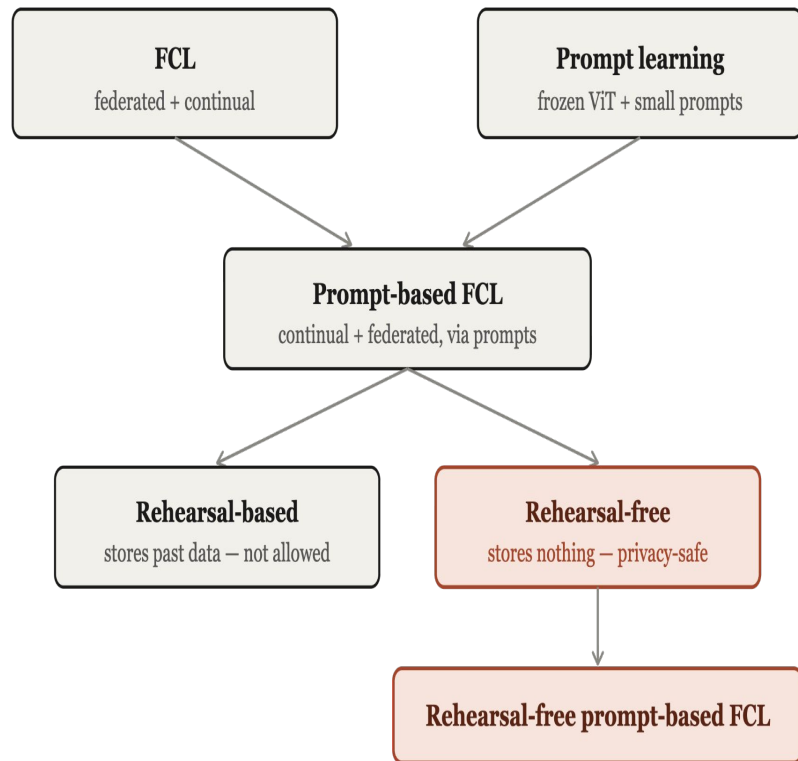
Introduction

- **Federated learning** - Many hospitals train one shared model without sharing raw patient data (privacy-preserving).
- **Continual learning** - the model keeps learning new diseases and imaging modalities over time, without forgetting old ones.
- Real medical AI needs **both at once** i.e. **Federated Continual Learning (FCL)**.



Introduction

- Preventing forgetting by storing and replaying past data (**rehearsal**) is **not allowed** as medical privacy forbids it.
- **Prompt learning** is the leading rehearsal-free approach: keep a large ViT **frozen**, train only small **prompts**.
- **Prompt-based FCL**: each client learns prompts locally, the server averages them.



Challenges

The accuracy failure - forgetting & drift

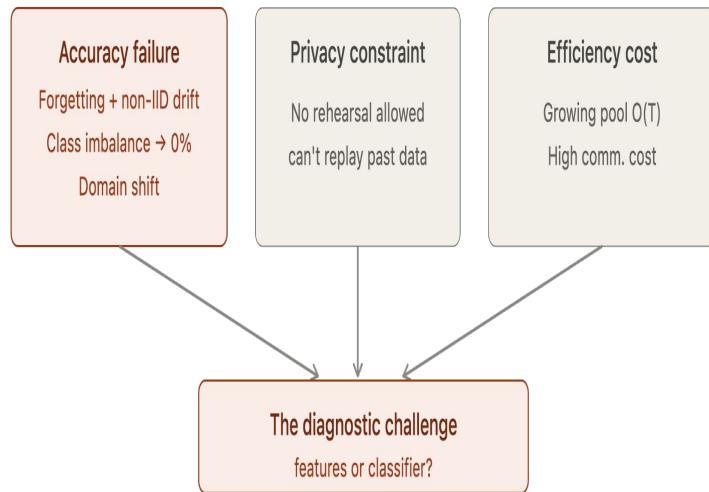
- **Catastrophic forgetting** - new tasks overwrite old ones; early tasks to $\sim 0\%$.
- **Non-IID clients** - hospitals hold skewed data; naïve FedAvg makes the shared classifier drift.
- **Class imbalance** - rare modalities (Derma, Retina) get dominated and collapses.
- **Domain shift** - medical scans are far out-of-distribution for ImageNet backbones.

The privacy constraint

- **No rehearsal allowed** - can't store or replay past patient data, removing the usual defence against forgetting.

The efficiency cost

- **Growing prompt memory** - the pool expands one prompt per task ($O(T)$); storage & communication grow without bound.
- **Communication cost** - prompts + classifier broadcast every round $\times$ every client strains real networks.



L2P^[1] - Learning to Prompt (CVPR 2022)

- First to bring prompt learning into continual learning.
- Keeps a pre-trained ViT completely frozen and learns a small prompt pool (key-value vectors).
- Uses **query-key similarity** to select the most relevant prompts for each input image.
- Achieves **task-agnostic continual learning** without storing old data or requiring task IDs during testing.

Limitation

Prompt selection is **discrete (non-differentiable)**, so gradients cannot optimize the selection process. It is designed for a **single centralized model** and does not support federated learning.

CODA^[2] -Prompt (CVPR 2023)

- Replaces L2P's **hard prompt selection** with a **weighted combination of prompts**.
- Weights come from attention over input keys, making the whole scheme **differentiable**.
- Uses **orthogonality** to reduce prompt interference and improve learning capacity.
- Outperforms L2P on rehearsal-free class- and domain-incremental benchmarks.

Limitation

The prompt pool grows with tasks (unbounded storage). Still **centralized** and evaluated only on natural images - untested in federation or on medical data.

REP^[3] - NeurIPS(2025)

- **Reduces the computation cost** of prompt-based continual learning.
- Introduces a **lightweight query** model to cheapen prompt selection.
- **Adds adaptive token reduction** and adaptive layer skipping to lower FLOPs and memory.
- **Preserves accuracy** while making prompt-based CL feasible on resource-constrained settings.

Limitation

Optimizes **local computation only**; does not reduce communication overhead. Designed for **single-device settings**, not federated learning.

PromptFL^[4] - IEEE(2024)

- Brings **prompts into federated learning** using a shared frozen foundation model (CLIP).
- Clients learn soft prompts locally instead of training the full network.
- Prompts are aggregated via FedAvg, drastically shrinking the communication payload.
- Reduces data requirement by exploiting the frozen foundation model.

Limitation

Not continual (assumes a fixed task set). Does not handle **catastrophic forgetting** or evolving prompt memory.

Related Work

Fed-CPrompt^[5] - ICML(2023)

- Federates prompt-based CL: a frozen ViT-B/16 extractor with a decomposed prompt pool.
- Each round, every client trains its current-task prompt + the classifier on local data.
- The server averages both prompts and classifier across clients (classical FedAvg).
- Enables rehearsal-free continual learning across private, non-IID clients.

Limitation

The prompt pool is unbounded, growing with every task.

Key Weakness

The trainable, federated-averaged classifier drifts and forgets the failure point our thesis fixes.

Method	Prompt-based	CL	FL	Bounded	Comm.
L2P [1]	✓	✓	—	—	—
CODA-Prompt [2]	✓	✓	—	—	—
REP [3]	✓	✓	—	—	✓ (compute)
PromptFL [4]	✓	—	✓	—	✓
Fed-CPrompt (base)	✓	✓	✓	—	—
Ours	✓	✓	✓	✓	✓

Research Gap

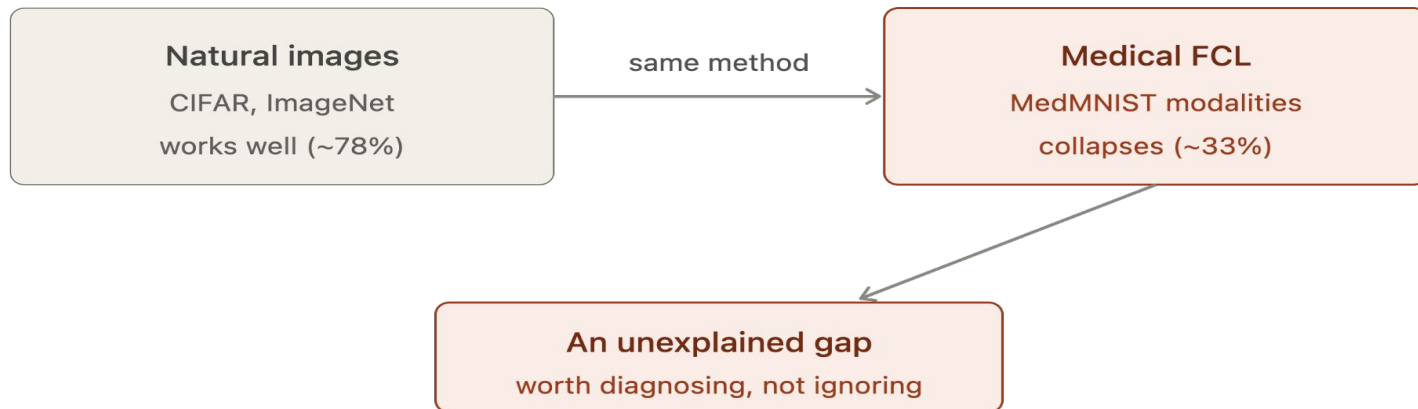
Existing prompt-based methods each solve part of the problem - none solve it together:

- **L2P / CODA-Prompt** - rehearsal-free continual learning, but **single-device** (no federation) and **only tested on natural images**.
- **REP** - makes prompting efficient, but **compute-only** and **single-device** (ignores communication and federation).
- **PromptFL** - brings prompts to federated learning, but **not continual** (assumes a fixed task set, no forgetting).
- **Fed-CPrompt** (base) - federated + continual, but the **prompt pool is unbounded** and the **classifier drifts** and it's **never been tested on medical data**.

The gap: no method is federated **and** continual **and** bounded **and** validated on medical imaging ,where privacy, class imbalance, and domain shift all appear at once.

Motivation

- Real medical AI must **keep learning** new diseases (continual) yet **can't share** patient data (federated) - FCL is its natural operating condition.
- Prompt-based CL (L2P, CODA-Prompt) fits medical privacy perfectly - no stored data, no model growth.
- **But it's only ever been tested on natural images** (CIFAR, ImageNet) and not tried on medical FCL.
- **So** accuracy stalls and under imbalance of small classes it collapses to 0%.
- A failure in this popular method is a real, unexplained gap worth diagnosing.

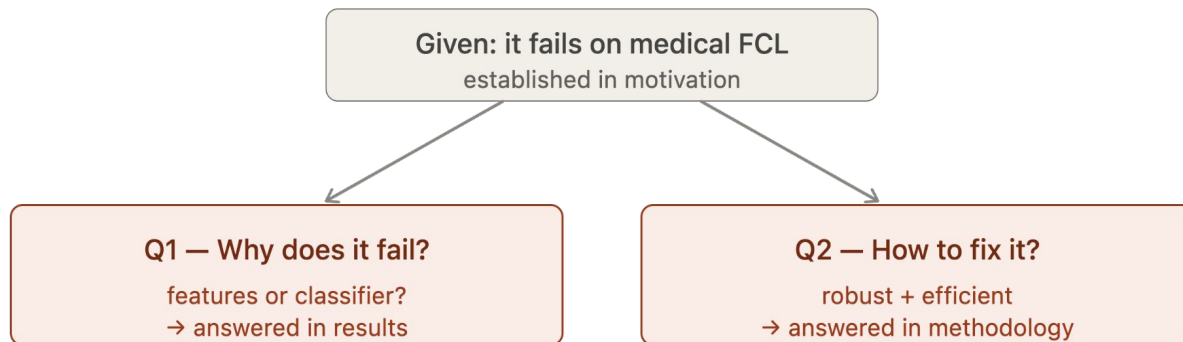


Problem Statement

Problem Statement

Given that prompt-based FCL fails on medical imaging, this work addresses **two questions**:

- **Why does it fail**, is the weakness in the frozen features, or in the trainable classifier?
- **How do we fix** it so the method is both robust (accurate under imbalance & non-IID, without forgetting) and efficient (bounded storage and communication)?

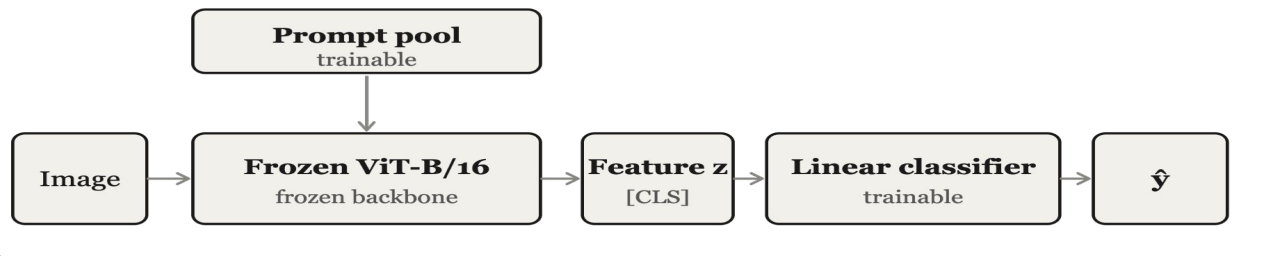


Base Paper (Fed-CPrompt) Architecture

10 federated clients — private, non-IID data



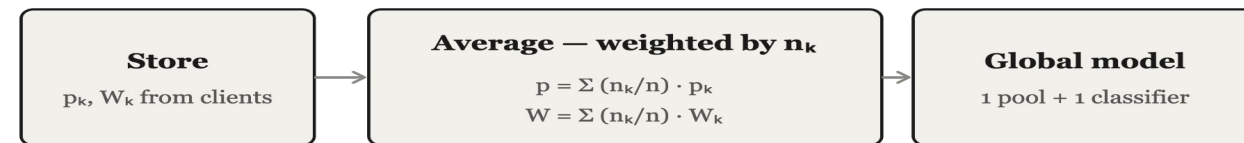
Local pipeline — identical on every client



① upload prompts + classifier

② broadcast global model

Server — FedAvg aggregation



Experimental Setup

Dataset - - MedMNIST (6 real imaging modalities: Path, Derma, Blood, OCT, OrganA, Retina)

- First 4 classes per modality - - shared **24-class** space (6 tasks $\times$ 4 classes)
- RGB, resized 28 $\rightarrow$ 224 to match ViT input

Two benchmarks, each isolating one failure mode

- **Balanced** - - 24 classes shuffled into 6 balanced tasks (~ 500 img/class) $\rightarrow$ isolates *non-IID clients*
- **Cross-domain** - - each task = one full modality, naturally imbalanced ($912 \rightarrow 87k$ images, $\sim 96\times$) - - isolates *class imbalance*

Federated partition - - 10 clients, image-level split; two regimes: **IID** (equal) and **non-IID** (Dirichlet $\beta = 0.5$), fixed seed

Model configuration

- **Backbone** - - frozen ViT-B/16 (ImageNet-21k), never trained
- **Prompts** - - CODA decomposed pool: 10 components/task, length 8, injected into blocks 0–4
- **Classifier** - - trainable linear head over all seen classes
- **Loss** - - Cross-Entropy + C2Loss ($\gamma = 0.5$, margin = 0.1, $\lambda = 0.1$)

Experimental Setup

Training protocol

- 40 communication rounds/task · 5 local epochs/round
- Adam, lr = 1e-4, batch 128, AMP · early stopping on 10% val
- Aggregation: FedAvg, weighted by client data size (n_i/n)

Reproduction check — CIFAR-100 (10×10) reproduces at **78.15%**, confirming a faithful baseline before medical testing.

Task	Modality	Classes	Train	Val	Test
1	PathMnist	0-3	35672	3964	3158
2	DermaMnist	4-7	1292	144	412
3	BloodMnist	8-11	5529	615	1758
4	OCTMnist	12-15	87729	9748	1000
5	OrganAMnist	16-19	5559	618	3398
6	RetinaMnist	20-23	912	102	380

Results

Baseline Results - Natural vs Medical (iid vs non-iid)

Benchmark	split	Final Acc(%)
CIFAR-100	iid	78.15
CIFAR-100	non iid	62.30
MedMNIST-balanced	iid	32.87
MedMNIST-balanced	non iid	26.48
MedMNIST-cross-domain	iid	28.97
MedMNIST-cross-domain	non iid	13.52

FINAL SUMMARY (label_skew, storage=unbounded)			
Task	Best Acc	Final Acc	Forgetting
1	74.92%	56.21%	18.71%
2	9.47%	0.24%	9.22%
3	0.00%	0.00%	0.00%
4	24.60%	24.50%	0.10%
5	0.18%	0.18%	0.00%
6	0.00%	0.00%	0.00%
AVG	18.19%	13.52%	4.67%

Base method (linear) fails on medical (as it tends to ~ 32.7)

Results

Baseline Results - Natural vs Medical (iid vs non-iid)

Benchmark	split	Final Acc(%)	Forgetting(%)
CIFAR-100	iid (cap=2)	76.2	8.21
CIFAR-100	non iid (cap=2)	62.04	13.60
MedMNIST-balanced	iid (cap=2)	32.54	17.72
MedMNIST-balanced	non iid (cap=2)	26.41	25.12
MedMNIST-cross-domain	iid (cap=2)	26.07	12.73
MedMNIST-cross-domain	non iid (cap=2)	15.39	21.94

Task	Best Acc	Final Acc	Forgetting
1	100.00%	11.27%	88.73%
2	49.75%	47.00%	2.75%
3	85.71%	73.98%	11.73%
4	27.60%	25.14%	2.46%
5	11.46%	10.84%	0.62%
6	27.99%	27.99%	0.00%
AVG	50.42%	32.70%	17.72%

MedMNIST results shows confirming catastrophic forgetting and we can say boundedness is allowed since fall in accuracy is nearly negligible

1. It Collapses on Medical Data

Applying the same pipeline to MedMNIST, accuracy drops to 32.70% (iid) and 26.48% (non-IID) - roughly half the natural-image performance. The method does not transfer to medical imaging.

2. Non-IID Clients Make It Worse

Moving from iid to non-IID, accuracy falls further (32.70 - $\rightarrow$ 26.48) and forgetting rises (17.72% $\rightarrow$ 25.12%). The federated-averaged classifier drifts as clients become heterogeneous.

3. Catastrophic Forgetting (Early Tasks)

Task 1 falls from $\sim$ 100% to $\sim$ 11% ($\sim$ 89% forgetting), while recent tasks stay high. This front-loaded pattern is the signature of recency bias in the shared classifier.

4. Boundedness of pool

Bounding the pool with random eviction (cap = 2) leaves accuracy essentially unchanged (unbounded 62.30 vs evict 62.04), proving growing prompt memory is not the cause.

Conclusion

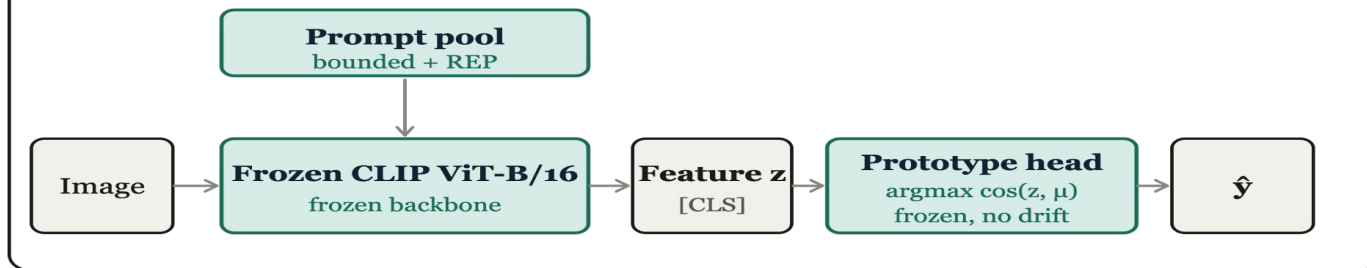
The base method fails on medical FCL because of catastrophic forgetting in the trainable, federated-averaged classifier (worsened by non-IID drift) - - not the frozen backbone or prompt pool.

Proposed Methodology

10 federated clients — private, non-IID data



Local pipeline — identical on every client



① upload prompts + class-mean sums

② broadcast global model

Server — aggregation (two paths)



Core idea

Keep the frozen prompt backbone, but replace the trainable, federated-averaged classifier — the part that drifts and forgets — with a frozen prototype head that is computed from features and never trained.

1. Feature Extraction

Image → frozen CLIP ViT-B/16
with bounded/REP prompts
injected into blocks 1–5 →
feature z (the [CLS] embedding).

**The backbone is never updated;
only the prompts are trainable.**

2. Prototype Head

Each class is one prototype = the
mean feature of its samples:

$$\mu_y = \text{mean of } z \text{ over class } y$$

3. Classification

Assign each feature to the nearest
prototype by cosine similarity:

$$\hat{y} = \underset{y}{\operatorname{argmax}} \cos(z, \mu_y)$$

4. Aggregation

The key distinction lies in the two-path aggregation process:

- **Prompts** - $\rightarrow$ FedAvg (gradient-averaged, weighted by data size).
- **Prototypes** - $\rightarrow$ class-mean sums.

5. Efficiency

Highly optimized communication and storage mechanism:

- **Bounded pool** (cap = c , random eviction) $\rightarrow$ constant storage/communication instead of $O(T)$.

6. Refresh

Ensuring representation consistency over time:

After all tasks, recompute every μ_y so old classes stay consistent with the current feature space.

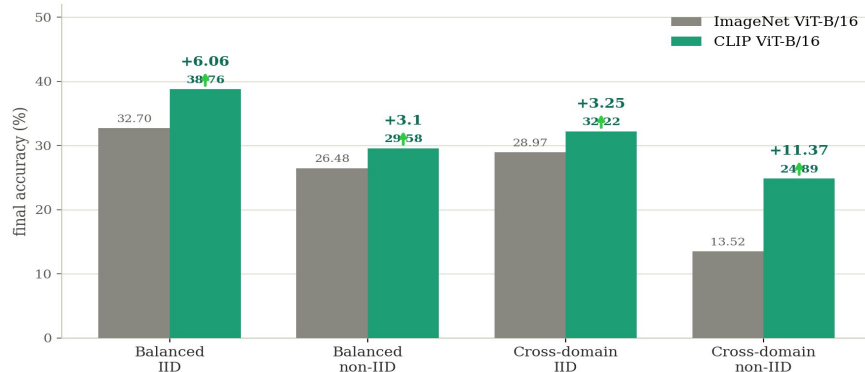
Implementation Results

Benchmark	iid (unbounded)	non iid (unbounded)
MedMNIST - balanced Linear head	38.76 , +6.06 ↑	29.58 , +3.1 ↑
MedMNIST - cross domain Linear head	32.22, +3.25 ↑	24.89, +11.37 ↑

```
=====
FINAL SUMMARY (iid, storage=unbounded)
=====
```

Task	Best Acc	Final Acc	Forgetting
1	98.84%	0.29%	98.55%
2	88.75%	35.50%	53.25%
3	94.39%	47.96%	46.43%
4	44.26%	37.43%	6.83%
5	70.28%	46.44%	23.84%
6	64.95%	64.95%	0.00%
AVG	76.91%	38.76%	38.15%

Backbone swap: ImageNet → CLIP raises accuracy in every setting



Implementation Results

Benchmark	iid (unbounded)	iid (cap=3)	iid (cap=2)	iid (cap=1)	non iid (unbounded)	non iid (cap=3)	non iid (cap=2)	non iid (cap=1)
MedMNIST (balanced) Prototype head	66.50	64.80	64.58	64.52	66.48	63.78	63.82	64.67
MedMNIST (cross domain) Prototype head	71.45	65.21	64.54	64.03	70.88	64.08	63.78	63.01

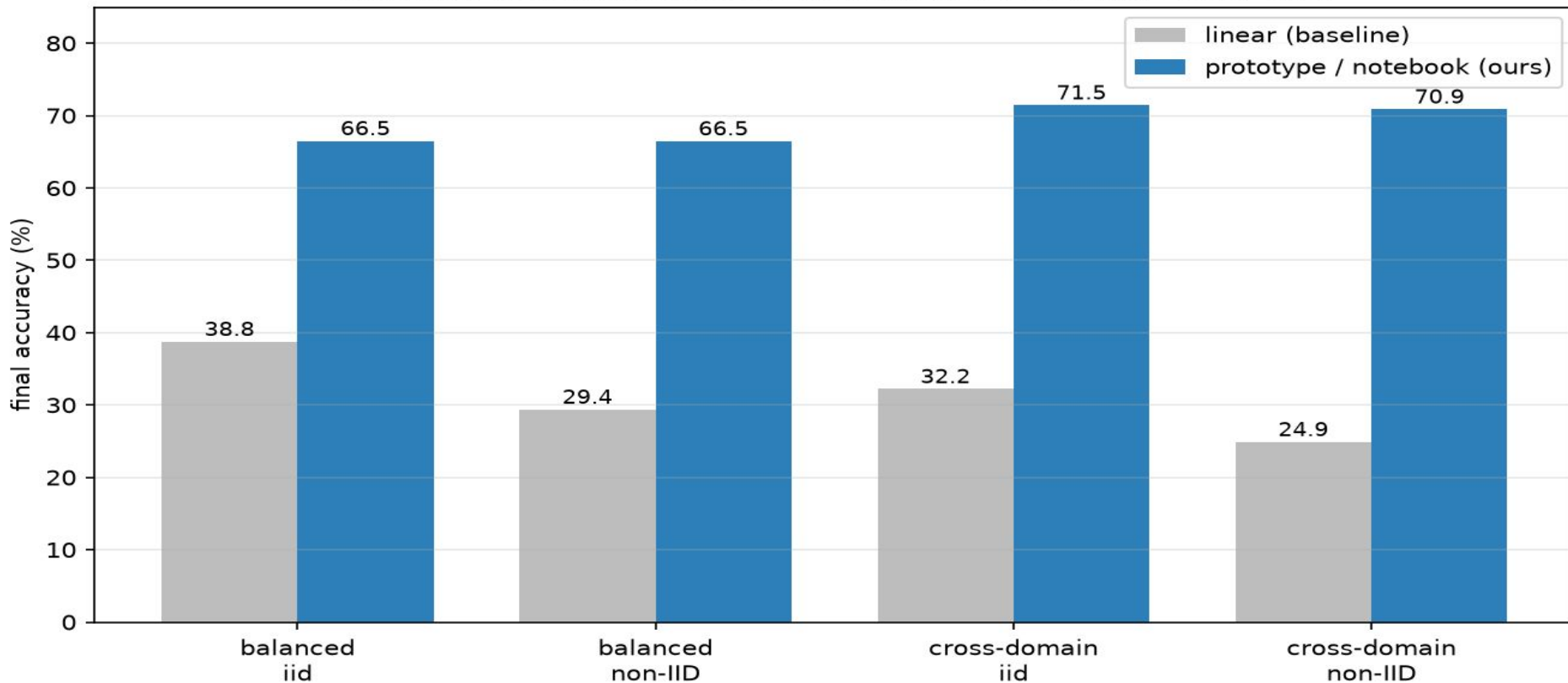
FINAL SUMMARY (iid, storage=unbounded)				
Task	Best Acc	Final Acc	Forgetting	
1	95.57%	94.81%	0.76%	
2	48.06%	46.84%	1.21%	
3	93.52%	89.14%	4.38%	
4	72.40%	67.00%	5.40%	
5	87.99%	88.55%	0.56%	
6	42.11%	42.37%	0.26%	
AVG	73.27%	71.45%	1.82%	

FINAL SUMMARY (iid, storage=cap=3/evict)				
Task	Best Acc	Final Acc	Forgetting	
1	96.30%	95.69%	0.60%	
2	48.06%	47.09%	0.97%	
3	93.52%	69.91%	23.61%	
4	72.80%	58.10%	14.70%	
5	90.61%	75.75%	14.86%	
6	45.00%	44.74%	0.26%	
AVG	74.38%	65.21%	9.17%	

FINAL SUMMARY (iid, storage=cap=2/evict)				
Task	Best Acc	Final Acc	Forgetting	
1	95.98%	94.93%	1.04%	
2	48.06%	44.90%	3.16%	
3	93.52%	69.17%	24.35%	
4	72.10%	58.90%	13.20%	
5	73.90%	74.07%	-0.18%	
6	43.68%	45.26%	-1.58%	
AVG	71.21%	64.54%	6.67%	

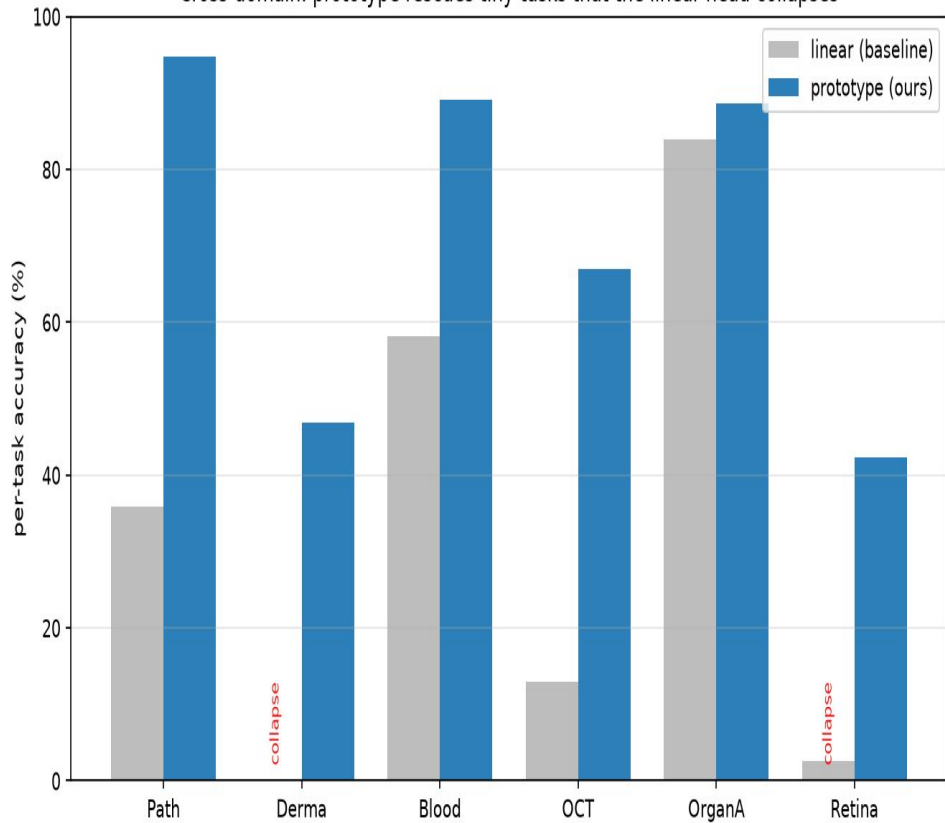
Implementation Results

Accuracy: prototype notebook vs linear baseline (MedMNIST, CLIP)

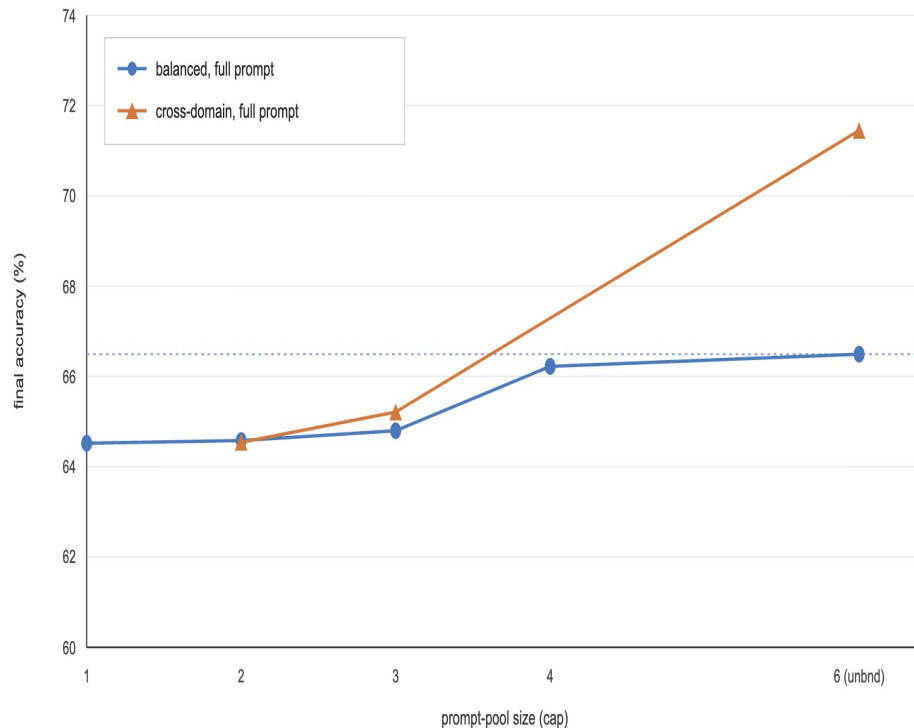


Implementation Results

Cross-domain: prototype rescues tiny tasks that the linear head collapses



Storage cap-sweep: bounding the pool is nearly free



Implementation Results (using REP)

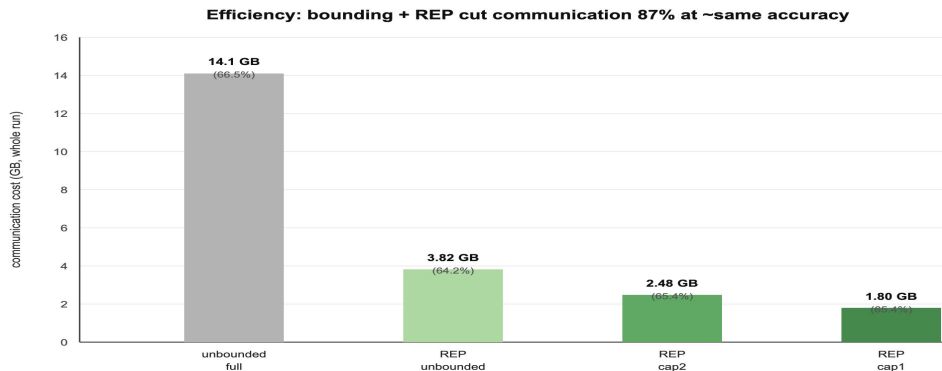
REP Config	Communication(in GB)
<i>without</i> REP, MedMNIST- balanced (Unbounded)	14.11
with REP, MedMNIST- balanced (Unbounded)	3.82
<i>without</i> REP, MedMNIST- balanced (Cap=2)	8.95
with REP, MedMNIST- balanced (Cap=2)	2.48
<i>without</i> REP, MedMNIST- balanced (Cap=1)	6.37
with REP, MedMNIST- balanced (Cap=1)	1.799

COST TABLE (prompt-pool storage + federated communication, REAL params)

clients=10 rounds/task=40 tasks=6 | full prompt=322,560 params (1.29 MB)
 REP lightweight prompt = 84,480 params (0.34 MB) = 26% of full

config	prompt(M,Lp)	pool	storage	comm	vs unbnd
unbounded (full)	(10,8)	6	7.74MB	14.112GB	0%
bounded (full)	(10,8)	1	1.29MB	6.370GB	55%
bounded + REP	(5,4)	1	0.34MB	1.799GB	87%

>> bounding saves : 7.741 GB (fewer prompts)
 >> REP saves : 4.571 GB (lighter prompts)
 >> bounded+REP : 1.799 GB vs unbounded 14.112 GB = 87% total reduction



Conclusion

- We diagnosed that the failure is not the features or the prompt pool, but catastrophic forgetting in the trainable, federated-averaged classifier.
- Our fix - a frozen prototype classifier that is never trained or averaged - nearly doubles accuracy ($38.8 \rightarrow >66.5$ balanced; $32.2 \rightarrow 71.5$ cross-domain).
- It is immune to non-IID clients (66.5% vs baseline 29.4%) and rescues imbalanced tasks (Derma 0 $\rightarrow$ 47%, Retina 2.6 $\rightarrow$ 42%).
- A bounded pool + REP lightweight prompts make it deployable - up to 87% less communication at $\sim 1\%$ accuracy cost.
- The result is a robust, efficient, and deployable FCL method for privacy-critical medical settings.

Future Work

- Reduce forgetting further: using much better prototypes instead of plain class-means.
- Reduce communication further: add prompt quantization/sparsification and utility-aware eviction on diverse streams.
- Raise accuracy: domain-appropriate (medical-pretrained) backbones and higher input resolution.
- Scale up: larger clinical images, and real multi-hospital deployment.

References

- [1] Z. Wang, Z. Zhang, C.-Y. Lee, H. Zhang, R. Sun, X. Ren, G. Su, V. Perot, J. Dy, and T. Pfister, "*Learning to prompt for continual learning*," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2022, pp. 139–149.
- [2] J. S. Smith, L. Karlinsky, V. Gutta, P. Cascante-Bonilla, D. Kim, A. Arbelle, R. Panda, R. Feris, and Z. Kira, "*CODA-Prompt: COntinual decomposed attention-based prompting for rehearsal-free continual learning*," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2023, pp. 11909–11919.
- [3] S. JeonJeon Ma, K. I. Kim, and M. Jeon, "*REP: Resource-Efficient Prompting for Rehearsal-Free Continual Learning*," arXiv preprint arXiv:2406.04772, 2024.
- [4] T. Guo, S. Guo, J. Wang, X. Tang, and W. Xu, "*PromptFL: Let federated participants cooperatively learn prompts instead of models — federated learning in age of foundation model*," IEEE Trans. Mobile Comput., vol. 23, no. 5, pp. 5179–5194, 2023.
- [5] G. Bagwe, X. YuFed-CPrompt and L. Zhang, "*Fed-CPrompt: Contrastive prompt for rehearsal-free federated continual learning*," in Federated Learning and Analytics in Practice (ICML Workshop), 2023. (arXiv:2307.04869)

Thank You!